Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	Integrated Preclinical Framework for Type 2 Diabetes with AI Analysis and Solvent-Free Formulations			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio (√), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI (√), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Bio Medical		Bio Medical	
	Medium Category	Small Category	Medium Category	Small Category
	Medical Bio	Small-Molecule Pharmaceuticals	Bio-processes/ Bio-Equipment	Bio-process Technologies
Concept of Proposed Technology	The proposed technology establishes an **integrated preclinical validation framework for Type 2 Diabetes** by structurally linking **AI-based metabolic pathway analysis** with **eco-friendly, solvent-free continuous extrusion–based formulation development**. Type 2 Diabetes is a complex metabolic disorder driven not only by peripheral insulin resistance but also by **disturbed energy balance and neuronal-metabolic crosstalk**. Preclinical research in this field often fails due to the low bioavailability of candidate materials and the lack of comprehensive evaluation systems that account for the **interaction between the central nervous system and glucose metabolism**. To address these limitations, the proposed framework applies a **solvent-free continuous extrusion process** to achieve high-dispersion solid formulations of poorly soluble anti-diabetic candidates. This approach improves physical stability and *in vivo* exposure. Crucially, the framework integrates this formulation technology with **Johns Hopkins' specialized expertise in energy balance and neuronal function**. Specifically, the project will utilize **specialized diabetes animal models (e.g.,**			

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	High-Fat Diet-induced insulin resistance models) to evaluate therapeutic efficacy on **glycemic control, insulin sensitivity, and glucose tolerance**. By analyzing the **molecular mechanisms of nutrient sensing (e.g., glucose signaling)**, the framework aims to identify and validate novel therapeutic targets for diabetes.
Rationale for Joint R&D Project	Diabetes is a multi-factorial disorder driven by the complex interplay between pancreatic dysfunction, systemic inflammation, and neuronal insulin resistance. Despite this complexity, traditional preclinical R&D remains overly "glucocentric," focusing on plasma glucose levels while neglecting the critical crosstalk between the central nervous system and peripheral metabolic organs. This narrow focus limits the translational success of new therapies. To address this, there is a clear need for an integrated framework that moves beyond standard models. A successful approach must bridge two critical gaps: • **Precision Delivery:** The ability to achieve controlled delivery of chemical entities—specifically distinguishing between central and peripheral targets—is essential to isolate and understand systemic vs. localized metabolic effects. • **Advanced Validation:** Formulation optimization for bioavailability must be linked to sophisticated *in vivo* validation using models that accurately simulate human neuro-metabolic impairment. By integrating site-specific delivery technology with high-fidelity physiological modeling, this initiative aims to validate therapeutics within a framework that reflects the true systemic nature of human diabetes. This approach ensures that candidate compounds are optimized not just for absorption, but for their ability to restore homeostasis across the entire metabolic axis.
Key Words	• Type 2 Diabetes • Insulin Resistance & Glucose Homeostasis • Neuronal-Metabolic Interaction • Solvent-free High-Dispersion Formulation • Nutrient Sensing & Energy Balance

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	**Start of the Project**	Level (3)	**End of the Project**	Level (6)
R&D Objectives	1. Identify key diabetic targets by analyzing nutrient sensing pathways and energy balance mechanisms (focusing on neuronal insulin resistance). 2. Translate bioavailability-enhanced formulations into effective preclinical study designs for anti-diabetic candidates. 3. Evaluate PK/PD and efficacy using specialized diabetes animal models (e.g., measuring glucose tolerance, insulin sensitivity). 4. Establish a continuous, reproducible preclinical development workflow for diabetes therapeutics.			
R&D Target Specifications	1. Reliable identification of drug targets related to improvement of impaired glucose metabolism and neuronal dysfunction in diabetes.			

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R&D Plan	2. Feasible implementation of preclinical studies using diabetes-induced animal models. 3. Demonstrated reproducibility and stability of solvent-free extrusion–based formulations. 4. Validation of therapeutic efficacy on glycemic control, and insulin signaling.
	1. Establish formulation conditions for anti-diabetic candidates using solvent-free continuous extrusion. 2. Conduct in vivo efficacy evaluation using diabetes models (Johns Hopkins). 3. Analyze molecular mechanisms regarding nutrient sensing, glucose signaling, and neuronal insulin resistance. 4. Consolidate results into a unified preclinical validation framework for diabetes.

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	• Provide and manage specialized animal models for Type 2 Diabetes (e.g., HFD-induced insulin resistance models). • Evaluate in vivo efficacy including Glucose Tolerance Tests (GTT), Insulin Tolerance Tests (ITT), glucose flux through hyperinsulinemic-euglycemic clamp and comprehensive metabolic phenotyping including food intake, energy expenditure and physical activities. • Investigate molecular mechanisms of glucose homeostasis in multi-organs, focusing on nutrient sensing (mTOR/AMPK pathways) and neuronal insulin resistance. • Provide scientific interpretation and advisory input for preclinical study design regarding metabolic regulations.
Expected R&R of Korean SME Partner	• Translate candidate materials into solvent-free extrusion–based preclinical formulations to enhance bioavailability. • Develop and validate formulation reproducibility and stability. • Conduct preclinical PK analysis. • Provide formulation, manufacturing, and validation infrastructure. • Coordinate technical integration across the preclinical workflow.

5. Expected Outcomes/Impact

Expected Outcomes and Impact	**[Technological Outcomes]** Establishment of an Integrated Preclinical Framework for Type 2 Diabetes: Development of a validation system linking AI-driven energy balance analysis (neuronal insulin resistance) with solvent-free formulation technology. Verification of Formulation Efficacy: Demonstration of enhanced bioavailability and therapeutic

	efficacy using High-Fat Diet (HFD) induced diabetes models. **[Commercialization and Expansion Potential]** Scalability to Metabolic Disorders: The framework can be expanded to other metabolic syndromes like obesity and MASLD, facilitating global technology transfer. Global Competitiveness: Strengthening the R&D capabilities of Korean SMEs through high-level validation with Johns Hopkins. **[Industrial and Economic Impact]** Paradigm Shift in Drug Development: Transitioning to mechanism-based preclinical R&D (focusing on central metabolic regulation) to reduce clinical failure risks. Bio–AI Ecosystem: Reinforcing a sustainable ecosystem by integrating AI analysis, advanced animal modeling, and formulation science.

6. References

(1) Arnold SE, **Kim SF**, et al. High fat diet produces brain insulin resistance, synaptodendritic abnormalities and altered behavior in mice. *Neurobiology of Disease*. 2014;67:79-87.

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(5) A review of hot-melt extrusion: process technology to pharmaceutical products. *Drug Development and Industrial Pharmacy*. 2012;38(5):557–583.